NTI

Academic Success & Intervention

Student Academic Health Analysis

A full end-to-end data analysis pipeline examining 12,156 students across five grade levels over a complete academic year.

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Program	**DEPI x NTI — Data Analysis Track**
Audience	**ENG. DONIA — Evaluation Committee**
Period	**March 2024 – March 2025**

SQL Server 2017 → Python 3.12 → Power BI

Pipeline Architecture

How raw data became actionable insight

01

Raw Data

5 CSV files

students · attendance · homework · performance · communication

→

02

SQL Server

Storytelling Phase

6-chapter narrative · Quality flags · Business questions

→

03

Python

Cleaning Phase

Normalization · Feature engineering · Export

→

04

Power BI

Dashboard Phase

6-page interactive dashboard · ENG. DONIA

Total Records Processed

12,156	364,680	60,780	36,468	24,312
Students	Attendance	Homework	Performance	Communication

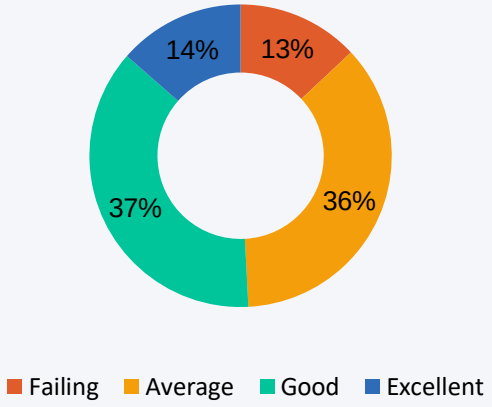
NTI Academic Success & Intervention

Executive Summary

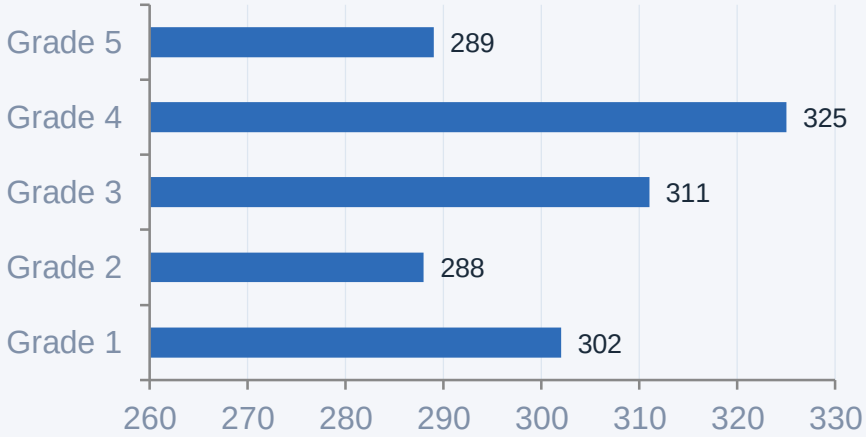
School health at a glance

12.156K	74.96	~2K	25.12%	49.99%
Total Enrollment	Avg Academic Score	At-Risk Students	Attendance Rate	HW Completion Rate

Score Band Distribution

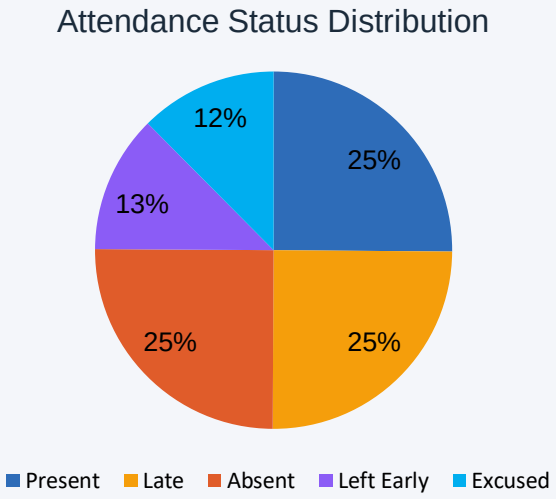


At-Risk Count by Grade



Attendance Analysis

Who is showing up, and when?

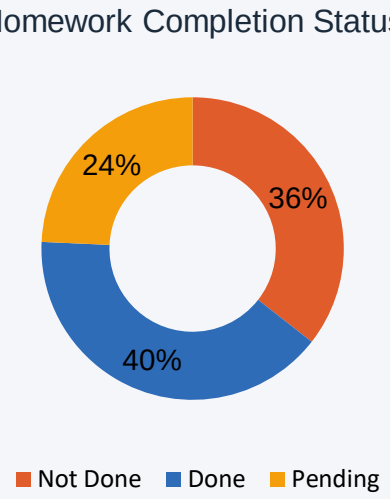


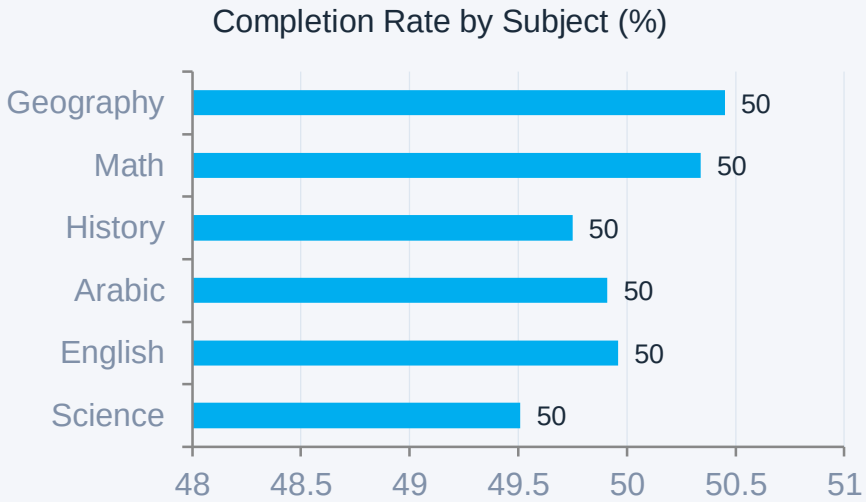
Key Finding

Only 25.12% of all records are marked Present — students are fully present in just 1 in 4 sessions. Late arrivals (24.96%) are as frequent as absences (~25%). **Absence is general, not subject-specific.**

Homework & Engagement

Are students doing the work?





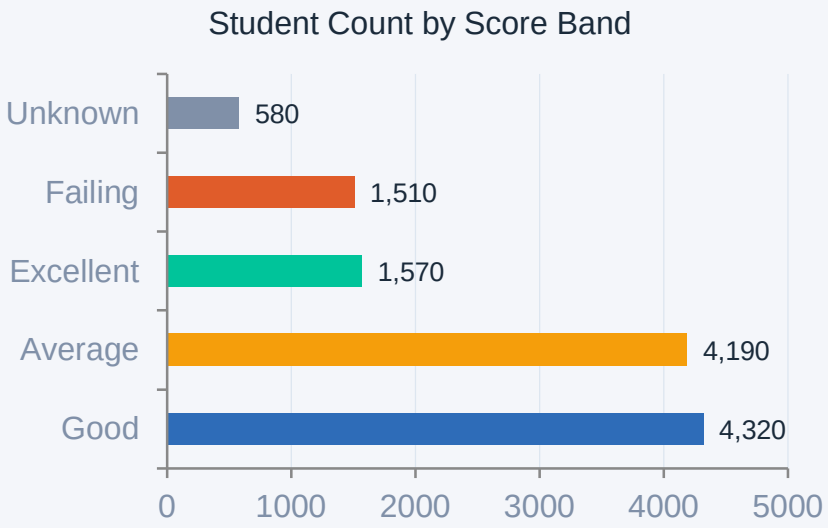
Key Finding

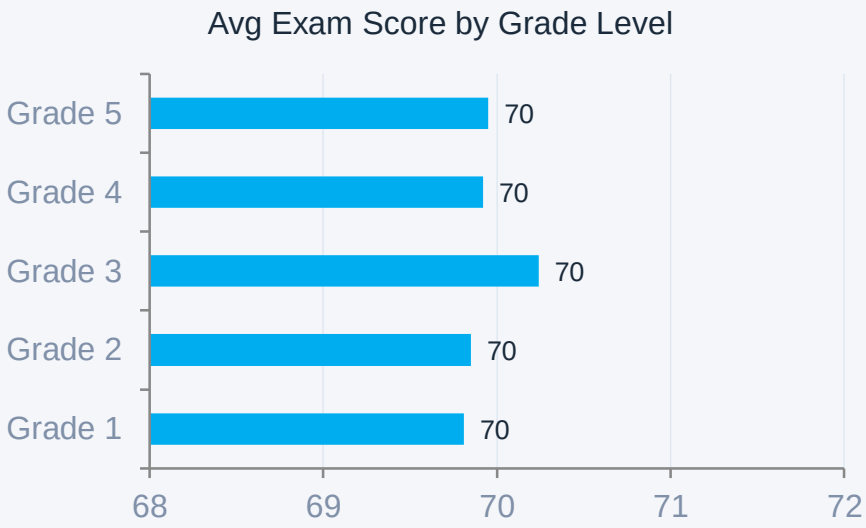
1 in 3 assignments is never completed. Science has the lowest completion rate. Guardian-signed assignments show marginally higher completion — suggesting parental involvement has some effect but is not the primary driver of engagement.

Academic Performance

How are students scoring?

74.96	40 – 110	12.87%	12.41%
School Avg Score	Score Range	High Achievers	Failing Rate





Parent Communication

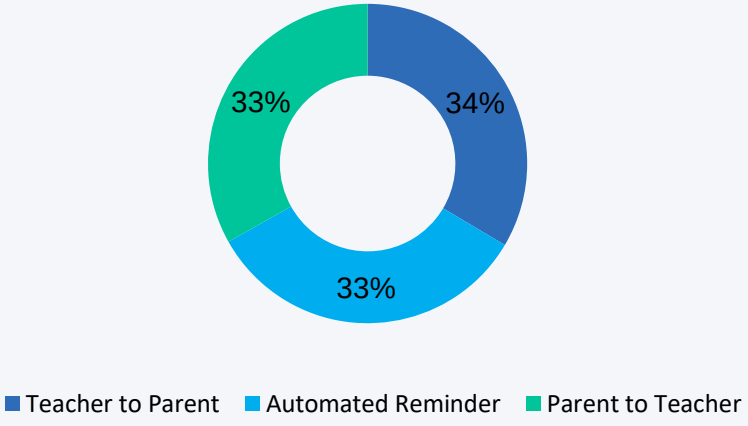
Is the school connected to parents?

24K
Total Messages

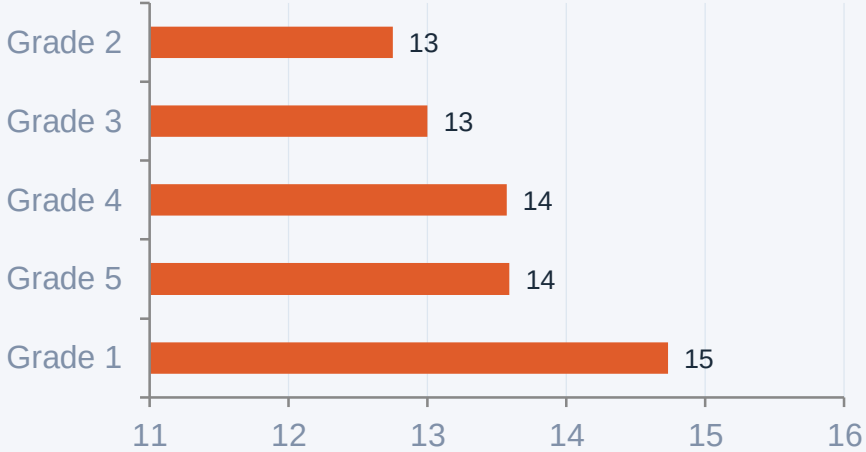
1,644
Silent Students

13.5%
Communication Gap

Message Type Distribution



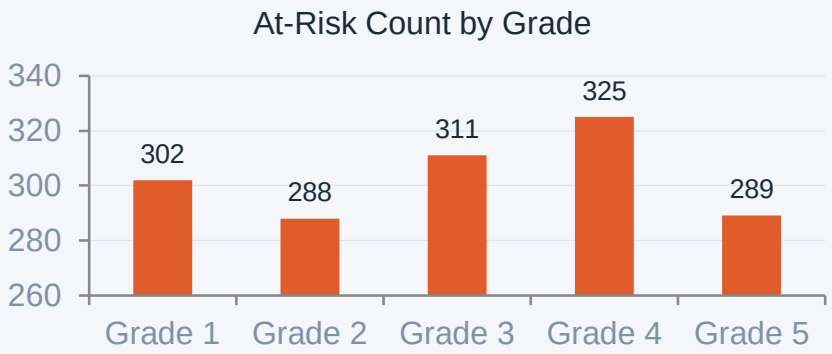
Silent Students % by Grade



At-Risk Intervention

Who needs action before the year ends?

~2K	12.41%	54.06	25.27%
At-Risk Students	% of School	At-Risk Avg Score	At-Risk Attendance Rate



At-Risk Flag Criteria

Attendance Rate	< 20%
Avg Exam Score	< 60
Missed Assignments	> 3

All 3 conditions must be true simultaneously

Most at-risk students have parent communication records. Communication is happening but not translating into improvement. The intervention strategy needs to change — not just its frequency.

Data Quality Issues & Fixes

What we found and how we resolved it

	Issue	Fix
Date_of_Birth Two date formats (ISO & US)	→	**pd.to_datetime(format='mixed')**
Attendance_Status Casing inconsistencies + 'absnt' typo	→	**str.upper() + map + ABSNT→ABSENT**
homework.Status Emoji + text variants mixed	→	**Set-based mapping → Done | Not Done | Pending**
Guardian_Signature Imported as BIT — blanks not NULL	→	**CAST in SQL + fillna(False) in Python**
Homework_Completion_% -5 invalid values + mixed % suffix	→	**Strip %, cast float, -5 → NaN, subject median fill**

Recommendations

What the data is telling us to do

01 Deploy the Intervention List

Use the At-Risk page as a live working tool. Each row is a student requiring a direct conversation before year-end.

02 Fix Grade 1 Communication Gap

14.73% of Grade 1 students have zero contact. Youngest students need the most parental engagement — build a structured onboarding protocol.

03 Rethink Communication Strategy

~200 at-risk students have active contact but declining performance. Volume is not the problem — quality and follow-through are.

04 Fix Data at Entry Point

Attendance typos, emoji status fields, and mixed date formats indicate no input validation. Standardize at source before the next year.

05 Build a Trend Detection Model

Add individual-level attendance trend monitoring. Flag students before they cross the at-risk threshold, not after.

06 Subject-Level Performance Model

Determine whether a student's low score is isolated to one subject or signals a broader academic decline across all subjects.

Thank You

NTI Academic Success & Intervention

Portfolio
nourhatem.netlify.app

LinkedIn
linkedin.com/in/nour-hatem-

Kaggle
kaggle.com/nourelhashash

GitHub
github.com/nour-hatem

12,156	~2K	6	3
Students	At-Risk	Dashboard Pages	Pipeline Phases